

Module 1

Numbers, Sets and Functions

1.1 Ordering and Inequalities

Question (1)

- (a) $x \in \left(-\infty, \frac{2}{3}\right] \cup (1, 2]$ (b) $x \in (-\infty, -4] \cup \left[-\frac{2}{3}, \infty\right)$
- (c) $x \in [-1, 3]$ (d) $x \in [1, 2) \cup (2, \infty)$ alternatively $[1, \infty) \setminus \{2\}$
- (e) $x \in \left[\frac{1}{3}, 7\right]$ (f) $x \in \left[-4 - \sqrt{15}, -\frac{3}{2}\right) \cup [-4 + \sqrt{15}, 2)$

1.2 The induction principle

Question (1)

Note: We will prove explicitly the expressions questions (a) and (c). Question (b) is similar and is left to the student. Question (d) is a conceptually difficult question. Its level of difficulty is much higher than that of assessable questions in Tests/Exam/Assignments. It is a question for students who enjoy to be challenged.

(a) First, we prove that $\mathcal{P}(n)$ is valid for $n = 1$. We have

$$LHS = \mathcal{P}(1) = 2 = 1 \cdot (1 + 1) = 2 = RHS.$$

Then, we state $\mathcal{P}(n+1)$, using the expression for $\mathcal{P}(n)$. On the LHS we add the extra term coming from the sum, i.e. $2(n+1)$. On the RHS we replace n with $(n+1)$. We have:

$$\mathcal{P}(n+1) : \quad \underbrace{2 + 4 + \dots + 2n}_{\text{LHS of } \mathcal{P}(n)} + 2(n+1) = \underbrace{(n+1)((n+1) + 1)}_{\text{RHS of } \mathcal{P}(n): n \rightarrow (n+1)} = (n+1)(n+2)$$

Now we use the assumption that $\mathcal{P}(n)$ is true. This implies that the LHS of $\mathcal{P}(n+1)$ can be rewritten using $\mathcal{P}(n)$, because it contains the sum of the first n even numbers:

$$\overbrace{2+4+\dots+2n}^{\text{RHS of } \mathcal{P}(n): n(n+1)} + 2(n+1) = n(n+1) + 2(n+1) = n^2 + 3n + 2.$$

This can now be proven to be equal to the RHS of $\mathcal{P}(n+1)$. In fact:

$$(n+1)(n+2) = n^2 + 2n + n + 2 = n^2 + 3n + 2.$$

In conclusion, we have proved that the LHS of $\mathcal{P}(n+1)$ is equal to the RHS of $\mathcal{P}(n+1)$, i.e. that $\mathcal{P}(n+1)$ is true if $\mathcal{P}(n)$ is true. The induction theorem allows us to state that the property is thus valid for all numbers $n \in \mathbb{N}$.

(c) Similarly to the previous case, we prove that $\mathcal{P}(n)$ is valid for $n = 1$. We have

$$\mathcal{P}(1): \quad 1 + 5 = 6 = (2+1)(1+1) = 3 \cdot 2 = 6.$$

Let us now state $\mathcal{P}(n+1)$, using the expression for $\mathcal{P}(n)$. On the LHS we add the extra term coming from the sum, i.e. $(4(n+1)+1) = (4n+5)$. On the RHS we replace n with $(n+1)$. We have:

$$\mathcal{P}(n+1): \quad \overbrace{1+5+9+\dots+(4n+1)}^{\text{LHS of } \mathcal{P}(n)} + (4n+5) = \overbrace{(2(n+1)+1)((n+1)+1)}^{\text{RHS of } \mathcal{P}(n): n \rightarrow (n+1)} = (2n+3)(n+2)$$

Now we use the assumption that $\mathcal{P}(n)$ is true. This implies that the LHS of $\mathcal{P}(n+1)$ can be written as:

$$\overbrace{1+5+9+\dots+(4n+5)}^{\text{RHS of } \mathcal{P}(n): (2n+1)(n+1)} + (4n+5) = (2n+1)(n+1) + (4n+5) = 2n^2 + 7n + 6.$$

As in the previous example, this is equal to the RHS of $\mathcal{P}(n+1)$, because

$$(2n+3)(n+2) = 2n^2 + 4n + 3n + 6 = 2n^2 + 7n + 6.$$

This completes our proof, since the LHS of $\mathcal{P}(n+1)$ is equal to the RHS of $\mathcal{P}(n+1)$.

(d) Here is a hint. Once you picture it, the question is much easier. Imagine taking a set with n elements and putting one extra element in it. The new set will certainly have all the subsets that the smaller set had. But it will have some new subsets too. In fact for each of the original subsets there will be exactly one new subset with the extra element added in. That means the number of subsets will double. You should now easily complete the proof.

Question (2)

We will prove inequalities in questions (a) and (b). Question (c) is slightly more difficult and left to the students who want to take the challenge.

(a) The first part of the question is straightforward;

$$\begin{aligned} \mathcal{P}(1) &: \quad 2^1 = 2 \geq 1 + 1 = 2, \\ \mathcal{P}(2) &: \quad 2^2 = 4 \geq 2 + 1 = 3, \\ \mathcal{P}(3) &: \quad 2^3 = 8 \geq 3 + 1 = 4. \end{aligned}$$

Let us now prove the proposition. First, $\mathcal{P}(1)$ is true, as shown above. We can now state $\mathcal{P}(n+1)$, using the expression for $\mathcal{P}(n)$. It is sufficient, in this case, to replace n with $(n+1)$ on both sides of the inequality:

$$\mathcal{P}(n+1) : \quad 2^{(n+1)} \geq (n+1) + 1 = (n+2)$$

Now we use the assumption that $\mathcal{P}(n)$ is true. If we rewrite $2^{(n+1)}$ as $2 \cdot 2^n$ and indicate the use of the inequality valid for $\mathcal{P}(n)$ with $*$, we have:

$$2^{(n+1)} = 2 \cdot 2^n \stackrel{*}{\geq} 2 \cdot (n+1) = 2n+2 > n+2,$$

given that $2n$ is always larger than n when $n \in \mathbb{N}$. In conclusion, we have proven above that (look at the first and last terms):

$$2^{(n+1)} > n+2.$$

Thanks to the induction theorem, we have proven that the inequality is valid for every $n \in \mathbb{N}$.

Note: The inequality can also be proven without using a proof by induction. In fact, taking the $\log_2$ for both sides of the inequality, we have:

$$2^n \geq (n+1) \quad \Rightarrow \quad n \geq \log_2(n+1)$$

Now, graphically, we can show that $x \geq \log_2(1+x)$ for $x \geq 1$, where $x \in \mathbb{R}$. This implies that $n \geq \log_2(n+1)$ for every $n \in \mathbb{N}$. Another possible proof is through direct application of the Bernoulli inequality.

Students are generally encouraged, when possible, to think about more than one route to the solution of any given problem.

(b) Again, it is straightforward to show that

$$\begin{aligned} \mathcal{P}(1) & : \quad 3^1 = 3 \geq 1^2 - 1 = 0, \\ \mathcal{P}(2) & : \quad 3^2 = 9 \geq 2^2 - 1 = 3, \\ \mathcal{P}(3) & : \quad 3^3 = 27 \geq 3^2 - 1 = 8. \end{aligned}$$

In analogy with the previous question, to prove the inequality we first state $\mathcal{P}(n+1)$:

$$\mathcal{P}(n+1) : \quad 3^{n+1} \geq (n+1)^2 - 1 = n^2 + 2n + 1 - 1 = n(n+2)$$

This is what we will need to prove.

Now, we use the assumption that $\mathcal{P}(n)$ is true. This case is similar to the previous one: if we rewrite $3^{(n+1)}$ as $3 \cdot 3^n$:

$$3^{(n+1)} = 3 \cdot 3^n \stackrel{*}{\geq} 3 \cdot (n^2 - 1) = 3n^2 - 3,$$

where $*$ denotes that we used the inequality valid for $\mathcal{P}(n)$. So, $3^{(n+1)} \geq 3n^2 - 3$. Does this imply that $3^{(n+1)} \geq n(n+2)$? In other words, is the following inequality true:

$$3n^2 - 3 > n(n+2) ?$$

Given that the above expression is a quadratic, we can solve it as usual, i.e.

$$3n^2 - 3 > n(n+2) \quad \Rightarrow \quad 2n^2 - 2n - 3 \geq 0$$

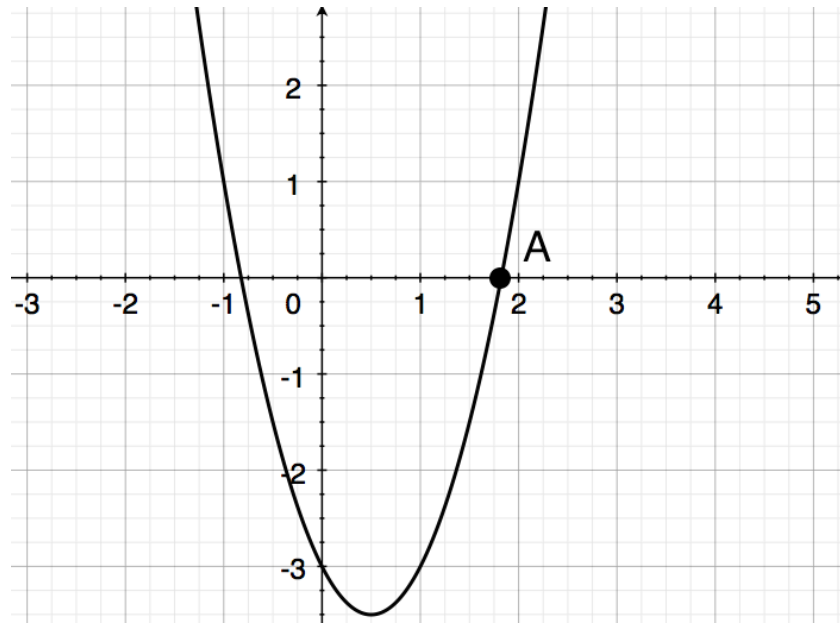
It is easy to see that the expression above is valid for $n \geq 2$, since

$$2x^2 - 2x - 3 \geq 0 \quad x \in (-\infty, (1 - \sqrt{7})/2) \cup ((1 + \sqrt{7})/2, \infty)$$

and $(1 + \sqrt{7})/2 \approx 1.83$ (see below). Since we are considering $\mathcal{P}(n+1)$ for large numbers n (certainly larger than 2), the inequality is true. In conclusion, overall we have proven that

$$3^{(n+1)} \geq 3n^2 - 3 > n(n+2),$$

hence verifying $\mathcal{P}(n+1)$.



The parabola corresponding to $y = 2x^2 - 2x - 3$. For any integer $n \geq 2$ we have $y > 0$.
Point A has coordinates $((1 + \sqrt{7})/2, 0)$

1.3 Domain and range of a function

Question (1)

- | | | | |
|-----|---|-----|--|
| (a) | $D = (-\infty, -1] \cup [1, \infty)$ or $\mathbb{R} \setminus (-1, 1)$, $R = \mathbb{R}$ | (b) | $D = \mathbb{R}$, $R = \left[-\frac{13}{4}, \infty\right)$ |
| (c) | $D = (-\infty, -\sqrt{3}) \cup (\sqrt{3}, \infty)$, $R = \mathbb{R}$ | (d) | $D = \mathbb{R} \setminus \{0\}$, $R = \mathbb{R}$ |
| (e) | $D = \mathbb{R} \setminus \{0\}$, $R = (0, \infty) \setminus \{1\}$ | (f) | $D = [-2, 3] \cup [7, \infty)$, $R = [0, \infty)$ |
| (g) | $D = (-\infty, -2] \cup [2, \infty)$, $R = [0, \infty)$ | (h) | $D = [-\pi, \pi] \setminus \left\{-\frac{\pi}{2}, 0, \frac{\pi}{2}\right\}$, $R = \mathbb{R}$ |
| (i) | $D = \mathbb{R} \setminus \{2\}$, $R = \mathbb{R} \setminus \{1\}$ | | |

1.4 Composition of functions

Question (1)

- (a) $f \circ g = f(g(x)) = \ln(\sqrt{x+3})$, $D_{f \circ g} = (-3, \infty)$
 $g \circ f = g(f(x)) = \sqrt{\ln x + 3}$, $D_{g \circ f} = [e^{-3}, \infty)$
- (b) $f \circ g = f(g(x)) = 2^{x^2+3x+1}$, $D_{f \circ g} = \mathbb{R}$
 $g \circ f = g(f(x)) = 2^{2x} + 3 \cdot 2^x + 1$, $D_{g \circ f} = \mathbb{R}$
- (c) $f \circ g = f(g(x)) = \sqrt{-x^4 - x^2 - 6}$, $D_{f \circ g} = \emptyset$
 $g \circ f = g(f(x)) = x^2 - x + 6$, $D_{g \circ f} = \mathbb{R}$
- (d) $f \circ g = f(g(x)) = \frac{3^x + 2}{3^x - 3}$, $D_{f \circ g} = \mathbb{R} \setminus \{1\}$
 $g \circ f = g(f(x)) = 3^{\frac{x+2}{x-3}}$, $D_{g \circ f} = \mathbb{R} \setminus \{3\}$

Question (2)

- (a) $h(h(x)) = 8x^4 - 24x^3 + 44x^2 - 39x + 24$, $D = \mathbb{R}$
- (b) $f(\sin(w(x))) = \frac{1}{\sin(x^2)}$, $D = \mathbb{R} \setminus \{\dots, -\sqrt{2\pi}, -\sqrt{\pi}, 0, \sqrt{\pi}, \sqrt{2\pi}, \dots\}$
- (c) $\ln(h(x)) = \ln(2x^2 - 3x + 4)$, $D = \mathbb{R}$
- (d) $f(\ln(x(x))) = \frac{1}{\ln(2x^2 - 3x + 4)}$, $D = \mathbb{R}$
- (e) $f(w(\sqrt{x} + 1)) = \frac{1}{(\sqrt{x} + 1)^2}$, $D = [0, \infty)$
- (f) $f(g(h(x))) = \frac{1}{(2x^2 - 3x + 4)^2 + 1}$, $D = \mathbb{R}$

Note: Question (d): the domain is $D = \mathbb{R}$ because $2x^2 - 3x + 4 > 0$.

1.5 Inverse functions

Question (1)

We indicate the inverse of the given (direct) function $y = f(x)$ as $f^{-1}(x)$. Also, D , R indicate the domain and the range, respectively.

- (a) $f^{-1}(x) = 1 + \sqrt{\ln x}$ $D_{f^{-1}} = [1, \infty)$, $R_{f^{-1}} = [1, \infty)$
- (b) $f^{-1}(x) = \sqrt[3]{1+x^2}$ $D_{f^{-1}} = [0, \infty)$, $R_{f^{-1}} = [1, \infty)$
- (c) $f^{-1}(x) = \exp(2^x - 1)$ $D_{f^{-1}} = \mathbb{R}$, $R_{f^{-1}} = (1/e, \infty)$
- (d) $f^{-1}(x) = (\log_2 x)^2$ $D_{f^{-1}} = [1, \infty)$, $R_{f^{-1}} = [0, \infty)$
- (e) $f^{-1}(x) = \sqrt{\frac{x+4}{1-x}}$ $D_{f^{-1}} = [-4, 1)$, $R_{f^{-1}} = [0, \infty)$

$$(f) \quad f^{-1}(x) = \frac{3 - \sqrt{1 + 4x}}{2}$$

$$D_{f^{-1}} = [-1/4, \infty), \quad R_{f^{-1}} = (-\infty, 3/2]$$

$$(g) \quad f^{-1}(x) = \frac{1 + \sqrt{3 - 2x}}{2}$$

$$D_{f^{-1}} = (-\infty, 3/2], \quad R_{f^{-1}} = [1/2, \infty)$$

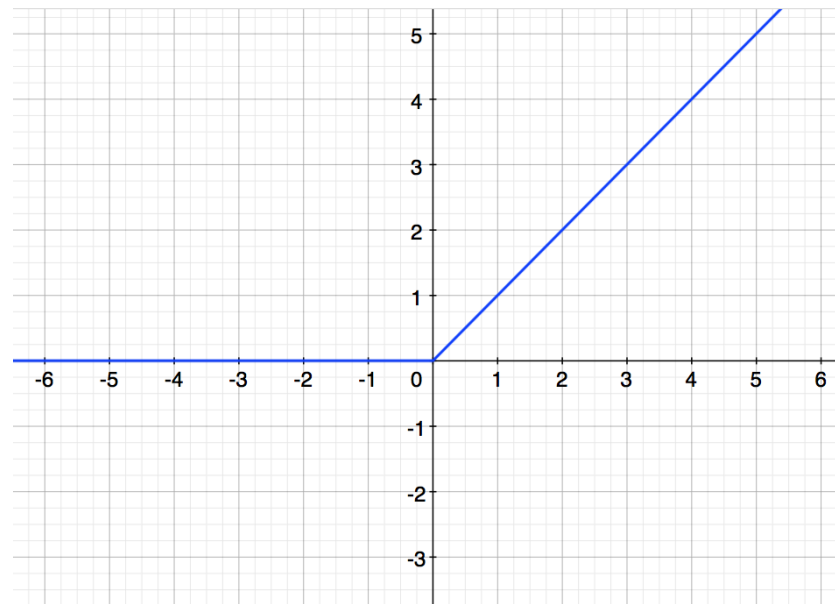
$$(h) \quad f^{-1}(x) = -\sqrt{x^2 - 1}$$

$$D_{f^{-1}} = [1, \infty), \quad R_{f^{-1}} = (-\infty, 0]$$

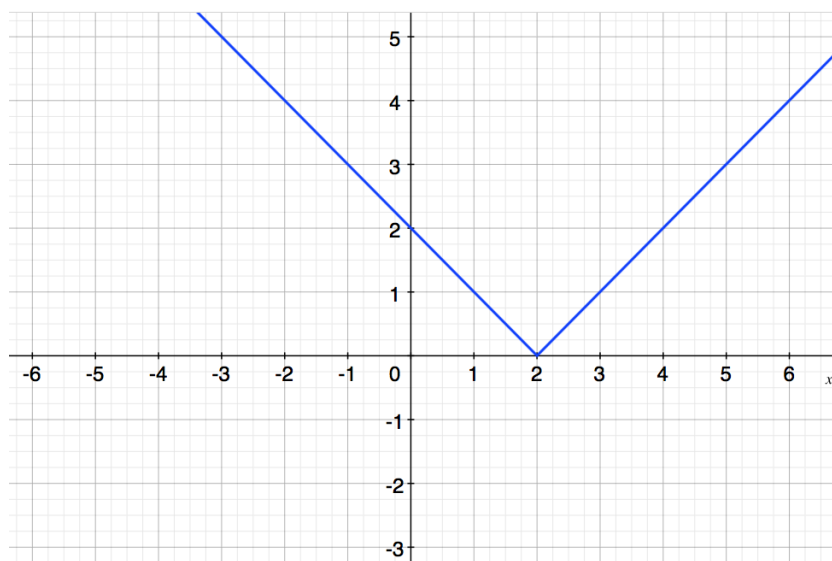
Note: For question (b), the natural domain of f^{-1} does not correspond to the chosen domain $D_{f^{-1}}$. We need to select a Domain for f^{-1} so that $D_{f^{-1}} = R_f$. Note that the natural domain of f^{-1} is instead $\mathbb{R}$.

1.6 Graphs of elementary functions

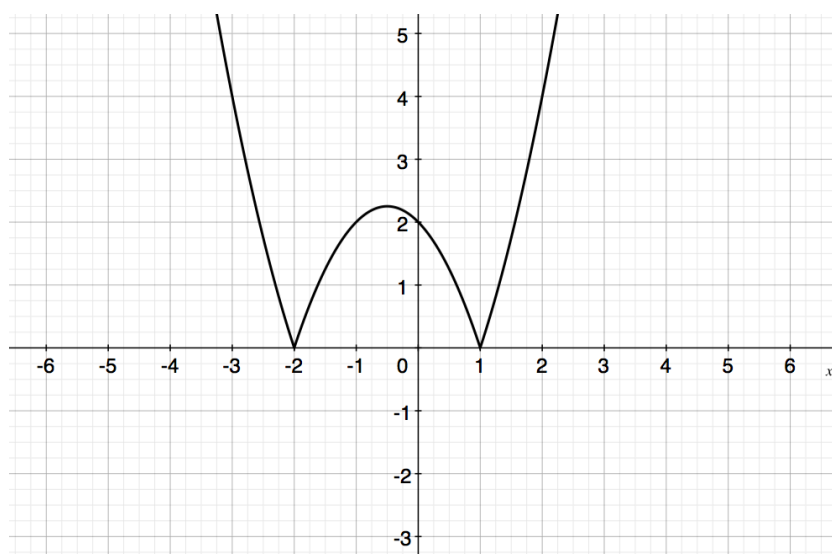
Question (1)



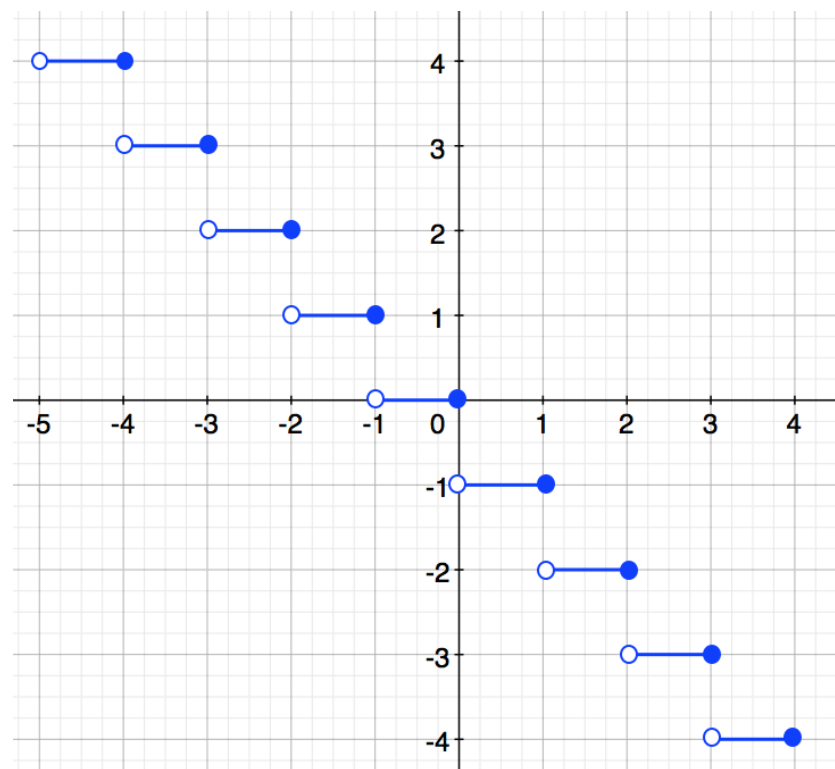
(a)



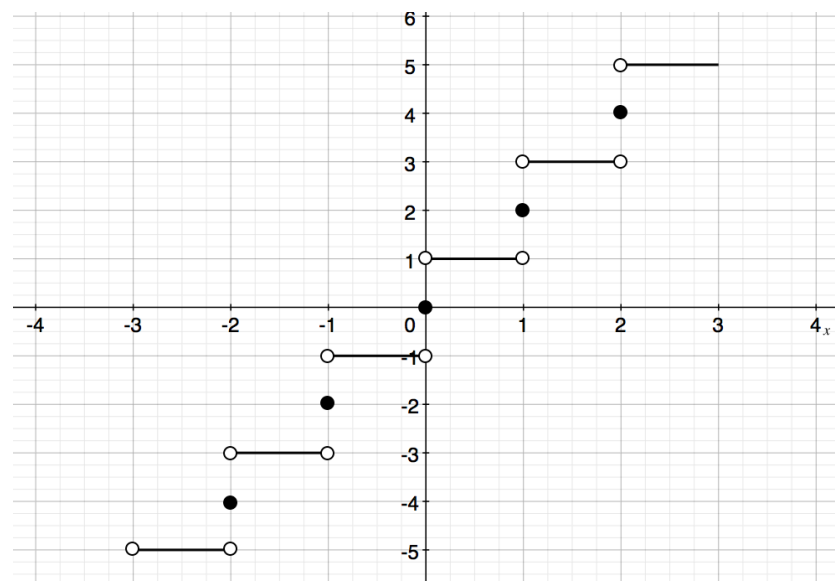
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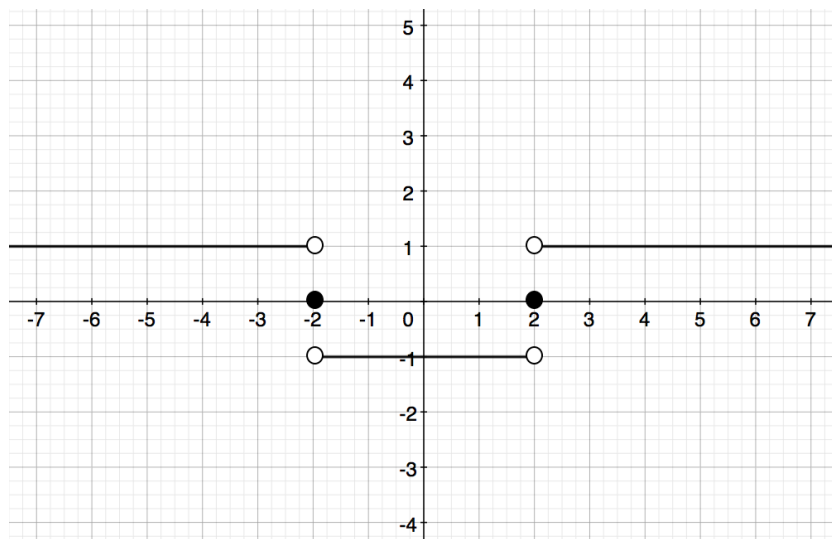
(c)



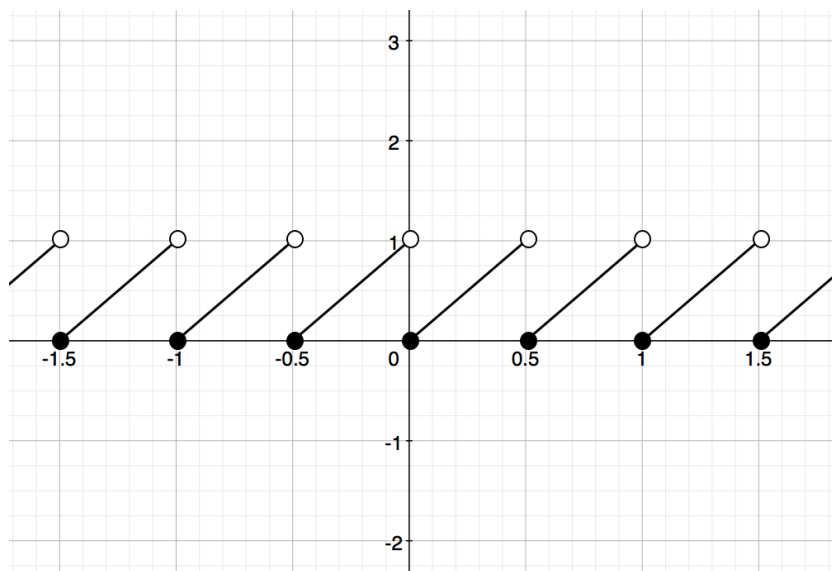
(d)



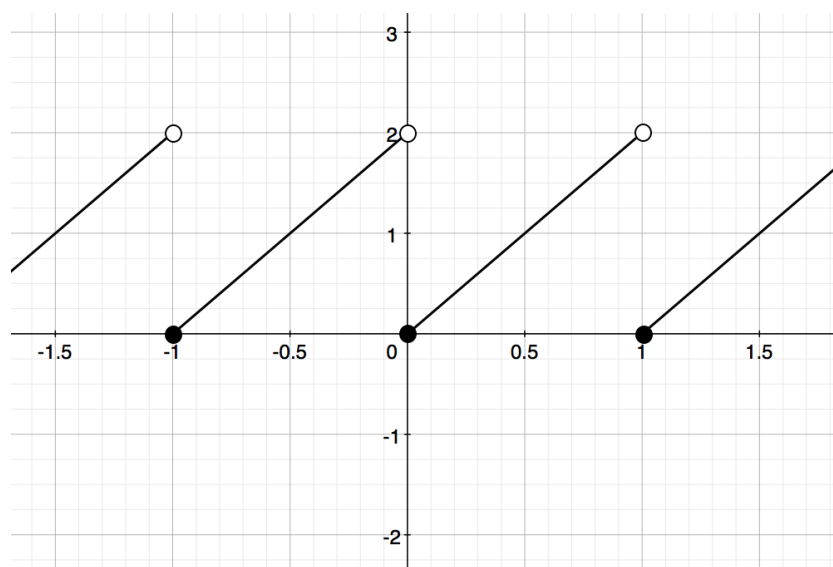
(e)



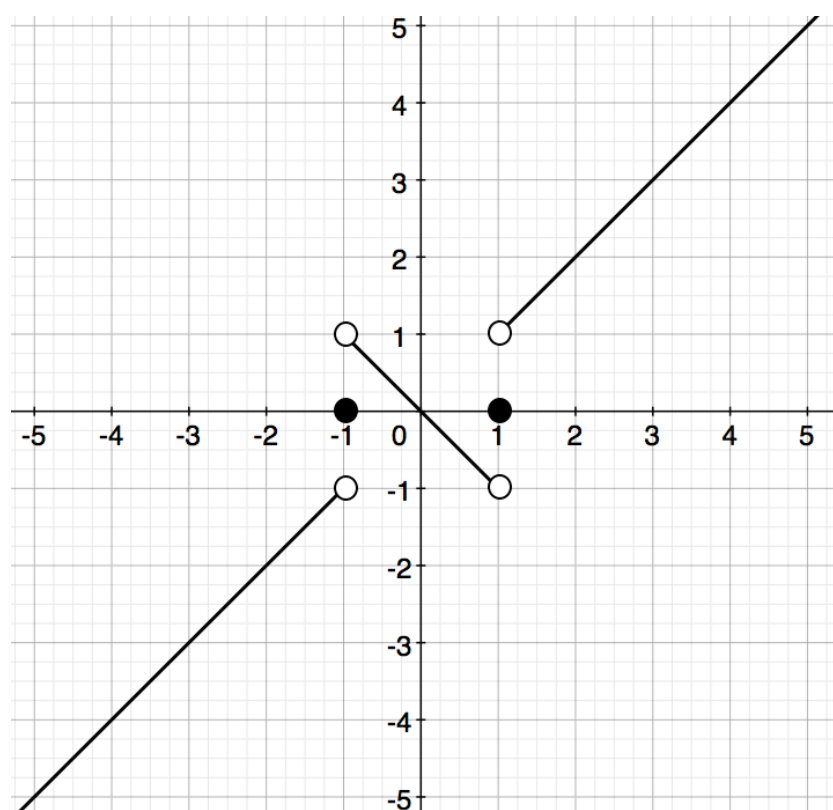
(f)



(g)



(h)



(i)

In parts (d) and (e) you might find the following helpful: Let $n \in \mathbb{N} \cup \{0\}$ i.e all natural numbers along with zero. Let $f \in (0, 1)$, i.e. all positive parts less than 1.

Then $\lfloor n \rfloor = n$, $\lfloor -n \rfloor = -n$, $\lfloor n + f \rfloor = n$ and
 $\lfloor -(n + f) \rfloor = \lfloor -n - f \rfloor = \lfloor -n - 1 + (1 - f) \rfloor = -n - 1$
 since $1 - f$ is also between 0 and 1.

1.7 Composition of trigonometric functions with their inverses

Question (1)

- | | | |
|-----------------------|---------------------------|---------------------------|
| (a) $\frac{\pi}{3}$ | (b) $\frac{2\pi}{3}$ | (c) $\frac{\pi}{3}$ |
| (d) $\frac{\pi}{4}$ | (e) $\frac{1}{2}$ | (f) $-\frac{24}{25}$ |
| (g) $\frac{161}{289}$ | (h) $\sqrt{\frac{6}{43}}$ | (i) $\frac{3}{\sqrt{34}}$ |

Question (2)

- | | | |
|------------------------------|--------------------------------|--|
| (a) $\frac{x}{\sqrt{1+x^2}}$ | (b) $\frac{\sqrt{1-x^2}}{x}$ | (c) $2x^2 - 1$ |
| (d) $\frac{1}{\sqrt{1+x^2}}$ | (e) $\frac{2x}{1+x^2}$ | (f) $\sqrt{1+x^2}$ |
| (g) $\frac{1}{x}$ | (h) $\sqrt{1 - \frac{1}{x^2}}$ | (i) $x\sqrt{1 - \frac{1}{x^2}} = \text{sign}(x)\sqrt{x^2 - 1}$ |

In part (i) we may take the x inside the square-root but we have to think first.

For negative x , $\text{arcsec}(x)$ is between $\frac{\pi}{2}$ and π . The tan of such angles is negative so we must retain the sign of x as it moves into the root.

1.8 Hyperbolic functions and their inverses

Question (1)

- | | | |
|--------------------|-------------------------|--|
| (a) $\frac{15}{8}$ | (b) $-\frac{4}{3}$ | (c) $\frac{-1/\sqrt{3} + \sqrt{3}}{1/\sqrt{3} + \sqrt{3}} = \frac{1}{2}$ |
| (d) $\frac{4}{5}$ | (e) $\frac{2}{e - 1/e}$ | (f) $\left(\frac{1 - e^2}{1 + e^2}\right)^2$ |

Note: in (e) and (f) do not use a calculator to evaluate because the question demands *exact* values.

Question (2)

- | | | |
|-------------------------------|---------------------------------|-------------------------------------|
| (a) $\frac{x^4 - 1}{x^4 + 1}$ | (b) $\frac{x(x - 2)}{2(1 - x)}$ | (c) $\frac{x^2 + 2x + 2}{2(1 + x)}$ |
|-------------------------------|---------------------------------|-------------------------------------|

Question (3)

- | | | |
|-------------------------------------|--------------------|-------------------------------------|
| (a) $\frac{3}{4}$ or $-\frac{3}{4}$ | (b) $\frac{17}{8}$ | (c) $\frac{3}{5}$ or $-\frac{3}{5}$ |
|-------------------------------------|--------------------|-------------------------------------|

Question (4)

(a) $\frac{\sqrt{29}}{5}$

(b) $-\frac{2}{\sqrt{29}}$

(c) $\frac{5}{\sqrt{29}}$

Question (5) Recall that Osborn's rule tells us to change the trig function to the corresponding hyperbolic function and at the same time introduce a minus sign on each term containing a pair of sine functions multiplied together. This includes squares of a sine or different sines in a product. A cube contains a square so gets a minus sign and a 4th power gets 2 sign changes etc. The rule also applies to sines disguised inside tans or cosecs etc.

(a) $\cosh^2(x) - \sinh^2(x) = 1$

(b) $\cosh^2(x) + \sinh^2(x) = \cosh(2x)$

(c) $\sinh(2x) = 2 \sinh(x) \cosh(x)$

(d) $1 - \tanh^2(x) = \operatorname{sech}^2(x)$

(e) $1 - \coth^2(x) = -\operatorname{cosech}^2(x)$

(f) $\sinh(x + y) = \sinh(x) \cosh(y) + \cosh(x) \sinh(y)$

(g) $\cosh(x - y) = \cosh(x) \cosh(y) - \sinh(x) \sinh(y)$

(h) $\tanh(x + y) = \frac{\tanh(x) + \tanh(y)}{1 + \tanh(x) \tanh(y)}$

(i) $\cosh(3x) = 4 \cosh^3(x) - 3 \cosh(x)$

(j) $\sinh(3x) = 3 \sinh(x) + 4 \sinh^3(x)$

(k) $\sinh(4x) = 4 \cosh(x)(\sinh(x) + 2 \sinh^3(x))$

(l) $\sinh(5x) = 5 \sinh(x) + 20 \sinh^3(x) + 16 \sinh^5(x)$

Question (6)

We will prove identities (a) and (c). Question (b) is similar. Question (d) is more challenging, and left to the student who enjoys challenging problems.

(a) Using the definitions of $\sinh$ and $\cosh$, the LHS becomes

$$\begin{aligned} \cosh(2x) - \sinh(2x) &= \frac{1}{2}(e^{2x} + e^{-2x}) - \frac{1}{2}(e^{2x} - e^{-2x}) \\ &= \frac{1}{2}\cancel{e^{2x}} + \frac{1}{2}e^{-2x} - \frac{1}{2}\cancel{e^{2x}} + \frac{1}{2}e^{-2x} = e^{-2x}. \end{aligned}$$

This is the desired result.

(c) For this case, we use the definitions of hyperbolic functions for the LHS:

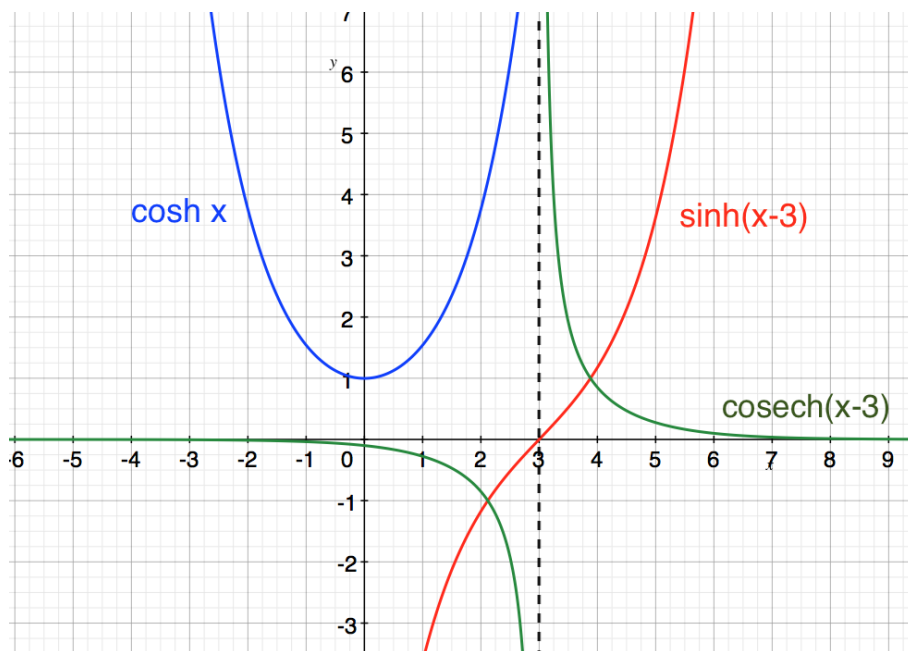
$$\begin{aligned} (\cosh x + \sinh x)^{100} &= \left(\frac{1}{2}(e^x + \cancel{e^{-x}} + e^x - \cancel{e^{-x}}) \right)^{100} \\ &= (e^x)^{100} = e^{100x} \end{aligned}$$

Then, we do the same for the RHS:

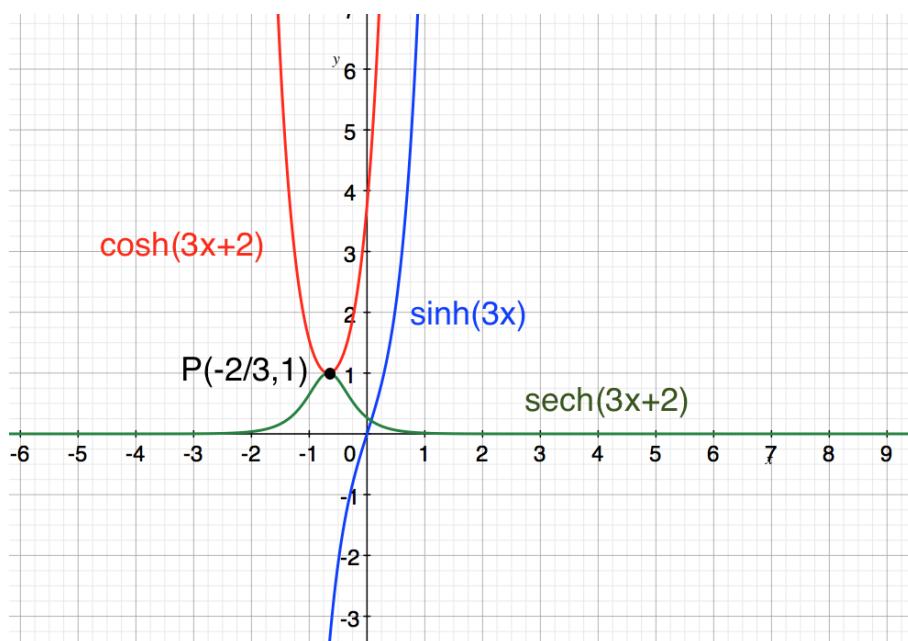
$$\begin{aligned}\cosh(100x) + \sinh(100x) &= \frac{1}{2}(e^{100x} + e^{-100x}) + \frac{1}{2}(e^{100x} - e^{-100x}) \\ &= \frac{1}{2}(e^{100x} + \cancel{e^{-100x}} + e^{100x} - \cancel{e^{-100x}}) = e^{100x}.\end{aligned}$$

This is the desired result.

Question (7a)



Question (7b)



Question (8)

(a) First, let us write down $\sinh(4x)$ explicitly:

$$\sinh(4x) = \frac{e^{4x} - e^{-4x}}{2}.$$

It helps to break the RHS into two terms. The first term is:

$$\begin{aligned} 4 \sinh x \cosh x &= 4 \left(\frac{e^x - e^{-x}}{2} \right) \left(\frac{e^x + e^{-x}}{2} \right) \\ &= (e^{2x} - e^{-2x}). \end{aligned}$$

We can use this result also to evaluate the second term on the RHS of our expression, i.e.

$$\begin{aligned} 8 \sinh^3 x \cosh x &= 2 \cdot 4 \sinh x \cosh x \cdot \sinh^2 x = 2 \cdot (e^{2x} - e^{-2x}) \frac{(e^x - e^{-x})^2}{4} \\ &= \frac{1}{2} (e^{2x} - e^{-2x}) (e^{2x} + e^{-2x} - 2) \\ &= \frac{1}{2} (e^{4x} + 1 - 2e^{2x} - 1 - e^{-4x} + 2e^{-2x}) \\ &= \frac{1}{2} (e^{4x} - e^{-4x}) - (e^{2x} - e^{-2x}). \end{aligned}$$

Summing the two terms, we have

$$\begin{aligned} 4 \sinh x \cosh x + 8 \sinh^3 x \cosh x &= (\cancel{e^{2x} - e^{-2x}}) + \frac{1}{2} (e^{4x} - e^{-4x}) - (\cancel{e^{2x} - e^{-2x}}) \\ &= \frac{1}{2} (e^{4x} - e^{-4x}) = \sinh(4x). \end{aligned}$$

(b) The procedure is similar and is left to the student.

Question (9)

Let us evaluate the expression explicitly:

$$\begin{aligned} \sinh^4 x &= \left(\frac{e^x - e^{-x}}{2} \right)^4 = \left(\frac{e^x - e^{-x}}{2} \right)^2 \left(\frac{e^x - e^{-x}}{2} \right)^2 \\ &= \frac{1}{16} (e^{2x} + e^{-2x} - 2)(e^{2x} + e^{-2x} - 2) \\ &= \frac{1}{16} (e^{4x} + 1 - 2e^{2x} + 1 + e^{-4x} - 2e^{-2x} - 2e^{2x} - 2e^{-2x} + 4) \\ &= \frac{1}{16} ((e^{4x} + e^{-4x}) - 4(e^{2x} + e^{-2x}) + 6) \\ &= \frac{1}{8} \left(\frac{e^{4x} + e^{-4x}}{2} - 4 \frac{e^{2x} + e^{-2x}}{2} + 3 \right) \\ &= \frac{\cosh(4x) - 4 \cosh(2x) + 3}{8} \end{aligned}$$

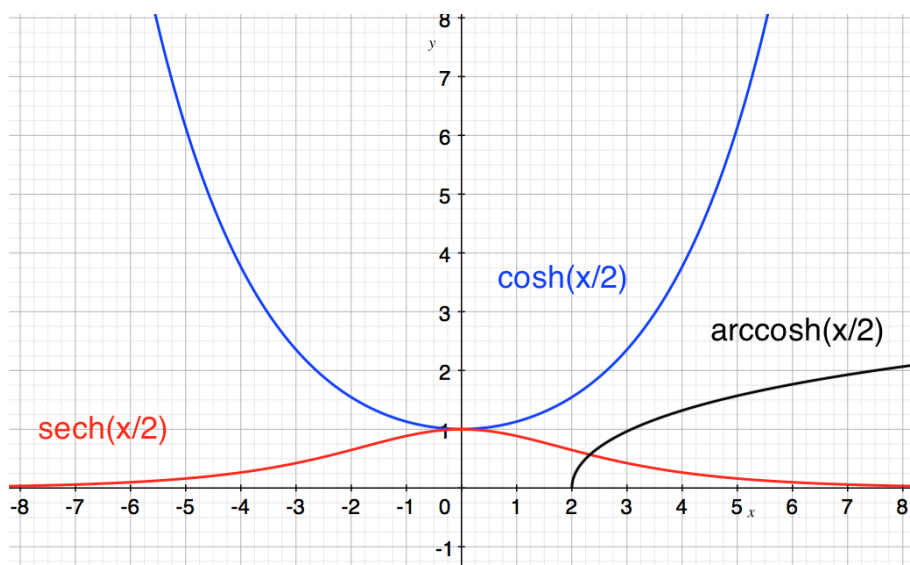
(b) The procedure is similar and is left to the student. Also note that the expression to prove in (b) is a consequence of the identity in question 8 (a) (see above).

Question (10)

- (a) $\sqrt{1+x^2}$ (b) $\frac{2x}{1-x^2}$
- (c) $\frac{1}{2x(16x^2-3)}$ (d) $\frac{1+x^2}{1-x^2}$

Note: For question (c), we have used the formula in question 8(b) (see above).

Question (11)



Question (12)

